

Impact of outsourcing on the supply chain strategy and processes of BMW, a German multinational automotive company

TABLE OF CONTENTS

1. Introduction	2
2. Background on Outsourcing in the Automotive industry	3
3. BMW's supply chain	4
3.1. Overview of BMW's supply chain and outsourcing practices	4
3.2. Factors driving BMW's outsourcing decisions	4
4. Analysis of BMW's outsourcing practices	5
4.1. Outsourcing strategy	5
4.2. Factors for supplier selection	5
4.3. Organisational structure	6
4.4. Business conduct	6
4.5. Operational activities	6
4.6. Business impact of outsourcing	8
4.7. Risks and challenges in outsourcing	9
5. Practical recommendations	10
6. Conclusion	11
7. References	12

1. Introduction

Global outsourcing has been a popular business strategy for many companies in recent years, as it enables them to take advantage of lower labour costs and other advantages in other countries while focusing on their core competencies. One of the companies that have successfully implemented global outsourcing in its operations is BMW.

BMW is a leading manufacturer of premium automobiles, motorcycles, and engines, and operates in over 150 countries. The company has a strong focus on innovation and quality, and

has implemented a number of global outsourcing strategies to enhance its operational efficiency and competitiveness.

The purpose of this report is to provide an in-depth analysis of BMW's outsourcing practices and its impact on the company's supply chain processes. The report will examine its outsourcing strategy and objectives, the identification of core competencies and non-core functions for outsourcing, supplier selection process and criteria, and review of BMW's supply chain structure and management. It will also assess the benefits and challenges of outsourcing for BMW, including risks such as supply chain disruptions and quality control issues. The report will conclude with recommendations for BMW and other companies in the automotive industry on effective outsourcing and supply chain management practices.

Overall, this report aims to provide insights into the outsourcing practices of BMW and their impact on the company's supply chain, as well as identify potential gaps and opportunities for improvement. The analysis will draw upon various models, frameworks, and data to provide a comprehensive understanding of BMW's outsourcing practices and their impact on the company's supply chain processes.

2. Background on Outsourcing in Automotive industry

Outsourcing has become a common practice in the automotive industry in recent years. It has been widely adopted by automotive manufacturers and suppliers to improve their competitiveness and reduce costs (Jacobides, MacDuffie and Tae, 2015). The automotive industry is one of the most globalised (Crossa, 2021) and complex industries in the world, and outsourcing has become an essential part of the industry's supply chain management.

Outsourcing in the automotive industry involves the transfer of some or all of the production processes, services, or components from an automotive manufacturer to a third-party supplier (Maltz and Ellram, 1999). These suppliers could be located locally or globally, and they may specialise in a specific area of the automotive industry, such as engine components, electronics, or interiors.

The main objective of outsourcing in the automotive industry is to improve efficiency, reduce costs, and enhance product quality. Outsourcing allows automotive manufacturers to focus on their core competencies, such as design and engineering, while outsourcing non-core functions to specialised suppliers. It also enables manufacturers to access the latest technologies and expertise of specialised suppliers, which they may not have in-house.

However, outsourcing also involves risks and challenges, such as supply chain disruptions, quality control issues, and legal challenges. These risks need to be managed effectively to ensure the success of outsourcing in the automotive industry.

3. BMW's supply chain

3.1 Overview of BMW's supply chain and outsourcing practices:

BMW has a complex supply chain that spans across several countries and involves numerous suppliers. The company's supply chain is divided into three tiers, with Tier 1 suppliers providing components directly to BMW, and Tier 2 and 3 suppliers providing components to Tier 1 suppliers (Karayel, 2017). BMW operates production facilities in several countries, including Germany, the United States, China, and South Africa, among others.

BMW outsources several components and processes, including engine and transmission manufacturing, electronics, and interior components. For instance, the company outsources engine and transmission manufacturing to suppliers such as Getrag and ZF Friedrichshafen. It also outsources the production of electronics to suppliers such as Bosch and Continental.

Additionally, BMW outsources the production of interior components, such as seats, to suppliers like Faurecia and Johnson Controls.

Cost reduction and specialisation are among the main factors driving BMW's outsourcing decisions. Outsourcing allows the company to reduce costs by taking advantage of economies of scale (Badorf et al., 2019) and lower labour costs in other countries. It also allows BMW to focus on its core competencies, such as design and marketing, while leaving manufacturing and production to specialised suppliers. Additionally, outsourcing provides BMW with flexibility in its production processes, as it can quickly adjust its supply chain to meet changing market demands.

3.2 Factors driving BMW's outsourcing decisions:

In terms of organisational structure, its outsourcing decisions are influenced by its focus on a decentralised organisational structure. The company's decentralised structure (Pries, 2009) allows for greater flexibility and agility in decision-making, as each division and region has greater autonomy in making its own sourcing decisions.

Overall, BMW's global outsourcing practices play a significant role in the company's operational activities and supply chain management. By outsourcing key components and processes, it can focus on its core competencies while also reducing costs and increasing flexibility.

4. Analysis of BMW's outsourcing practices

4.1 Outsourcing strategy:

BMW's outsourcing strategy focuses on leveraging the expertise and resources of external suppliers to improve efficiency, reduce costs, and enhance the quality of its products and services. BMW's primary outsourcing objectives include improving supply chain efficiency, enhancing product quality, reducing costs, and fostering innovation and flexibility (Delic and Eysers, 2020).

BMW has identified its core competencies to be engineering, design, and marketing. Non-core functions such as manufacturing, logistics, and IT are often outsourced to specialised third-party providers. This allows BMW to focus on its core competencies while leveraging the expertise of external providers for non-core functions.

S. No.	Core Competencies	Non-Core Functions
1.	Design and engineering	Production of standard components
2.	Brand management	Manufacturing of low-volume parts
3.	Marketing and sales	Logistics and transportation
4.	Quality control and testing	IT support and maintenance

Table 1: Examples of core competencies & non-core functions at BMW

4.2 Factors for supplier selection:

BMW's supplier selection process involves a thorough evaluation of potential suppliers based on their capabilities, capacity, quality, delivery reliability, financial stability, and cost competitiveness. BMW also places a strong emphasis on sustainability and ethical standards in its supplier selection process. Description for such criteria of selection is provided in table 2.

Criteria	Description
Quality	Suppliers must meet strict quality standards and have a track record of producing high-quality products

Cost	Suppliers must be competitive in terms of pricing, and offer cost savings over in-house production
Delivery Time	Suppliers must have the ability to deliver products on time and meet production schedules
Capacity	Suppliers must have the capacity to meet BMW's production needs
Sustainability	Suppliers must meet BMW's environmental and social responsibility standards

Table 2: Some of the supplier selection criteria

4.3 Organisational structure:

One of the key drivers of BMW's global outsourcing activities is its organisational structure, which is designed to enable the company to be agile and responsive (Ambe and Badenhorst-Weiss, 2010) to changing market conditions. The company has a decentralised structure, which allows it to operate in different regions with different cultural and business practices. This structure enables BMW to take advantage of local knowledge and expertise, while still maintaining its global brand identity.

4.4 Business Conduct:

BMW's business conduct is guided by its Code of Conduct, which outlines the company's commitment to ethical behaviour, environmental sustainability, and social responsibility. The company expects its suppliers to adhere to these principles as well.

4.5 Operational Activities:

BMW's outsourcing strategy has enabled it to optimise its operations and achieve its business objectives. The company has implemented several models, including the Just-in-Time (JIT) model (Battini, Boysen and Emde, 2012), to reduce its inventory and improve its operational efficiency. The JIT model allows BMW to maintain a low inventory level and receive parts from suppliers only when they are needed. This has reduced BMW's storage costs and improved its cash flow.

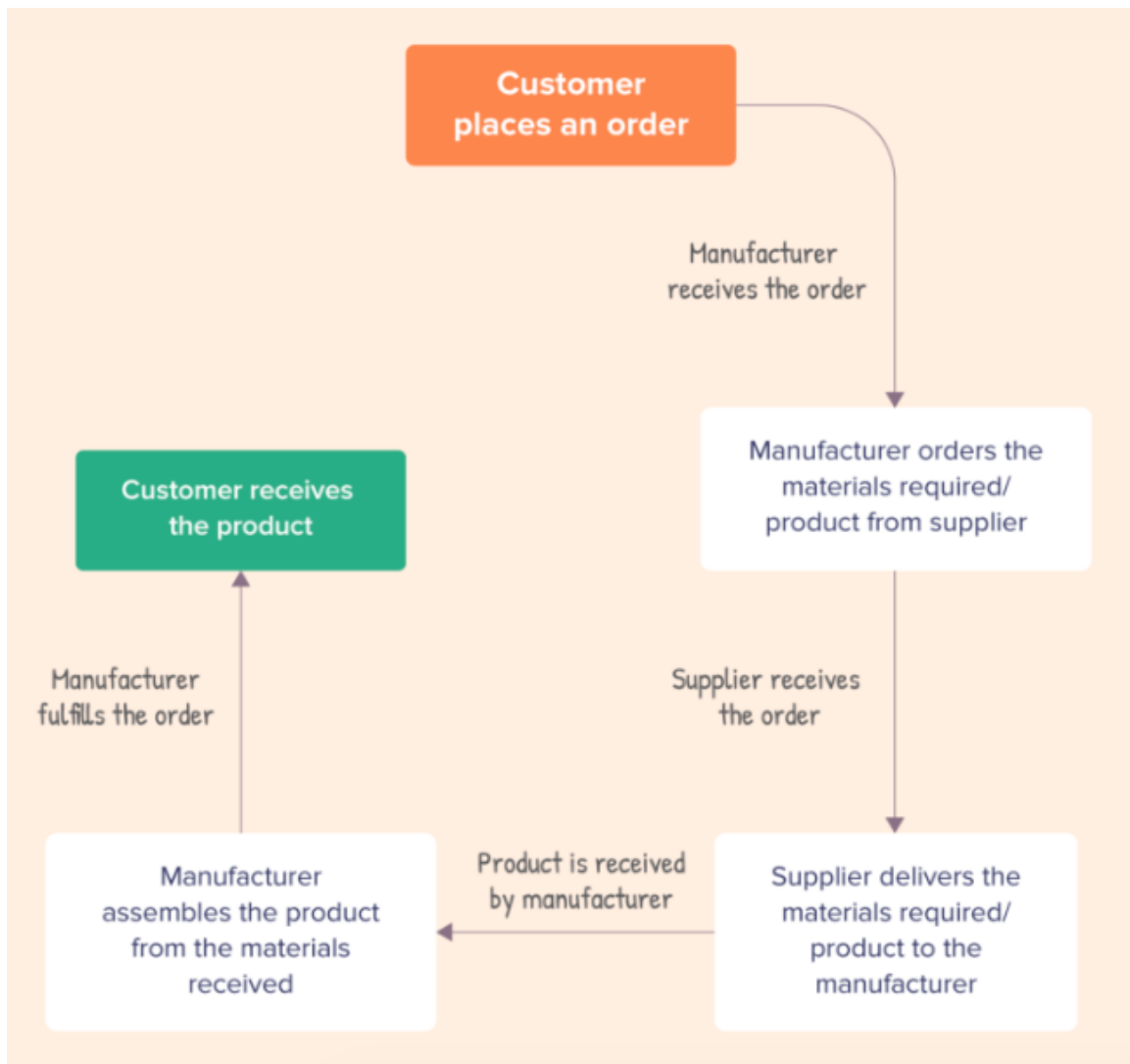


Figure 1: JIT model, Zoho (Kesavan, 2019)

In terms of operational activities, BMW has implemented a number of outsourcing strategies to enhance its manufacturing and distribution processes. For example, the company has outsourced the production of certain components to suppliers in countries such as China and Mexico, where labour costs are lower. This has helped to reduce production costs and improve the overall efficiency of the manufacturing process.

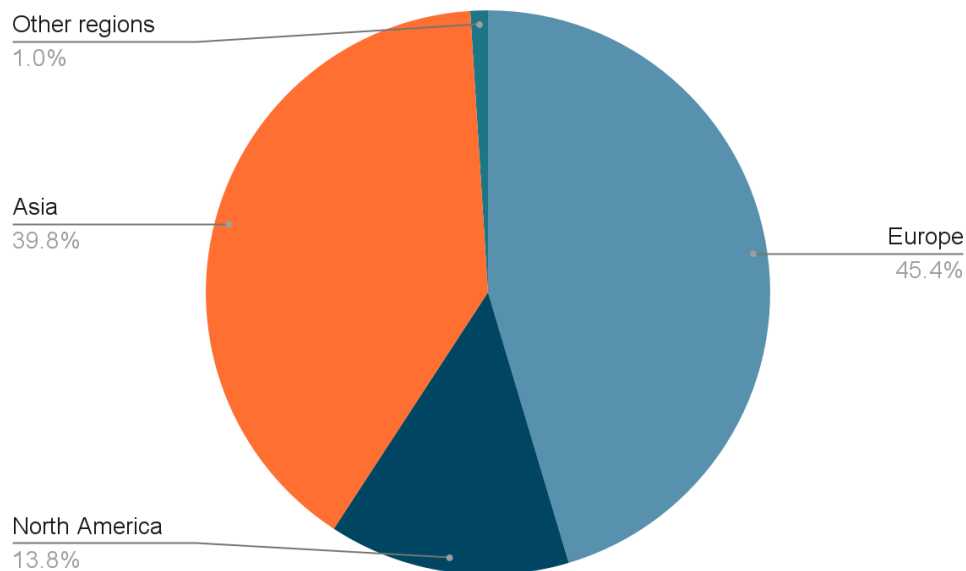


Figure 2: BMW's Global production by region (Supplementary information on agenda items 9.1 to 9.6: Consolidated Financial Statements and Management Report of BMW AG and BMW Group as of 31 December 2021, 2021)

4.6 Business impact of outsourcing:

Cost savings achieved through outsourcing: According to BMW's annual report for 2020 (OUR RESPONSIBILITY. OUR FUTURE, 2020), the company reduced its purchasing costs by €1.3 billion through measures including "optimization of supplier management, improved purchasing conditions and effects of outsourcing measures."

Supplier diversity: In 2020, BMW had around 13,000 direct suppliers and approximately 30,000 indirect suppliers globally, according to the company's sustainability report. This suggests a strategy of working with a large number of suppliers in order to diversify risk and ensure a reliable supply chain.

Quality control: BMW has implemented various quality control measures to ensure that outsourced components meet the company's standards. For example, the company has a Supplier Quality Excellence Program that evaluates suppliers based on factors such as delivery reliability, quality management, and sustainability. In 2020, 67% of BMW's direct suppliers achieved the highest rating in this program.

Innovation: Outsourcing can provide access to specialised expertise and technologies that a company may not possess in-house. BMW has formed partnerships with suppliers (Rhodes, Warren and Carter, 2006) in order to develop new technologies for its vehicles, such as advanced driver assistance systems and electric drivetrain components.

Risk management: Outsourcing can introduce new risks to a company's supply chain (Thun and Hoenig, 2011), such as supplier bankruptcy or quality issues. BMW has implemented various measures to manage these risks, such as supplier evaluations and regular audits of supplier facilities. The company also maintains a risk management system that includes monitoring of potential risks in the supply chain.

4.7 Risks and challenges in outsourcing:

Outsourcing also brings some risks and challenges. Supply chain disruptions can occur due to various reasons, such as natural disasters, political instability, or supplier bankruptcy. For example, in 2011, flooding in Thailand (Haraguchi and Lall, 2015) disrupted the supply chain for BMW's electronic components. This led to a shortage of components, which caused production to slow down and increased costs. As a result, BMW had to find alternative suppliers and invest in supply chain risk management.

Another significant risk associated with outsourcing is quality control issues. As BMW relies on external suppliers to produce components and products, it can be challenging to maintain consistent quality standards. For example, if a supplier fails to meet quality standards, this could lead to delays in production or even safety concerns for customers. To mitigate this risk, BMW works closely with its suppliers to ensure they comply with quality standards and implements a robust quality control process.

Apart from this as BMW operates with multiple outsourcing vendors in different countries, there are several legal challenges that the company may face. Some of the key legal challenges are:

1. Compliance with local laws: When outsourcing to different countries, BMW must comply with the local laws of each country. This includes labour laws, tax laws, and environmental regulations. Failure to comply with these laws can result in legal penalties, reputational damage, and loss of business.
2. Intellectual property rights: Outsourcing can involve sharing sensitive information and proprietary technology with third-party vendors. BMW must ensure that its intellectual property rights are protected and that the vendors are not using its technology or designs for their own purposes or sharing them with competitors.

3. Contractual disputes: Outsourcing relationships are governed by contracts, and disputes can arise if one party fails to meet its obligations or if there are differences in interpretation. It is important for BMW to have clear and well-defined contracts with its outsourcing vendors that outline the rights and responsibilities of each party.
4. Political instability: Outsourcing to countries with political instability can be risky, as changes in government or civil unrest can disrupt supply chains and operations. BMW must monitor political developments in the countries where it operates and have contingency plans in place to deal with potential disruptions.
5. Corruption: Outsourcing to countries with high levels of corruption can also be a risk, as the company may be exposed to illegal or unethical practices.

To address these challenges, it is important for BMW to conduct due diligence on its outsourcing vendors and ensure that they are operating ethically and in compliance with local laws & it should work closely with local govt.

5. Practical recommendations

After conducting a thorough analysis of BMW's outsourcing practices and supply chain management, there are some areas where BMW can improve its outsourcing strategy (Sturgeon, Van Biesebroeck and Gereffi, 2008). The following are some gaps in outsourcing for BMW and recommendations for BMW and other companies in the automotive industry regarding outsourcing and supply chain management.

Gaps in Outsourcing for BMW:

1. Limited Supplier Diversity: BMW has a limited number of outsourcing partners, which can increase the risk of supply chain disruption. It is recommended that BMW expands its supplier base to include more diverse and reliable partners, especially in emerging markets.
2. Lack of Transparency: There is a lack of transparency in BMW's supply chain, which can make it difficult to identify potential issues and address them proactively. It is recommended that BMW implements transparency measures, such as supplier audits, to ensure compliance and visibility across the supply chain.
3. Limited Focus on Sustainability: While BMW has made some progress towards sustainability in its supply chain, there is still room for improvement. It is recommended

that BMW works closely with its suppliers to improve their sustainability practices, especially in areas such as resource management and carbon footprint reduction.

4. **Insufficient Risk Management:** BMW needs to enhance its risk management strategies to prevent potential supply chain disruptions. This can be achieved by establishing contingency plans, diversifying suppliers, and investing in technology to monitor supply chain activities and detect any issues early on.

Recommendations for BMW and other companies in the automotive industry:

1. **Collaborative Partnerships:** Companies in the automotive industry should focus on developing collaborative partnerships with their outsourcing partners to ensure a seamless supply chain. This can involve joint planning, communication, and risk management efforts to ensure that both parties benefit from the partnership.
2. **Digitalization:** The use of technology, such as blockchain (Fraga-Lamas and Fernandez-Carames, 2019) and the Internet of Things (IoT) (Stankovic, 2014), can improve supply chain transparency and traceability. Companies should invest in these technologies to enhance their supply chain management and reduce the risk of disruptions.
3. **Supplier Diversity:** Companies should diversify their supplier base to reduce the risk of supply chain disruption. This can involve identifying new suppliers in emerging markets or developing relationships with smaller, local suppliers.
4. **Sustainability:** Companies should prioritise sustainability in their outsourcing practices and work closely with their suppliers to improve sustainability practices. This can involve setting sustainability goals, implementing green initiatives, and monitoring suppliers' sustainability performance.

To address these legal challenges, BMW must have a robust legal and compliance program in place. This program should include policies and procedures for selecting and monitoring outsourcing vendors, as well as training for employees on legal and ethical issues related to outsourcing. BMW should also work closely with its outsourcing vendors to ensure that they are complying with all relevant laws and regulations.

6. Conclusion

In conclusion, BMW's outsourcing strategy has played a significant role in its supply chain and operational efficiency, allowing the company to focus on its core competencies while leveraging

the capabilities of external partners. By outsourcing non-core functions and leveraging the expertise of its partners, BMW has been able to reduce costs, improve quality, and increase flexibility in its supply chain.

However, there are still gaps in BMW's outsourcing strategy that can be addressed to further improve its supply chain management. These include a need for greater transparency and collaboration with outsourcing partners, a more comprehensive risk management strategy, and a focus on sustainability and social responsibility in its outsourcing relationships.

Overall, the automotive industry continues to face challenges and opportunities in outsourcing and supply chain management, and BMW serves as an example of a company that has successfully navigated these complexities. By continuing to prioritise strategic outsourcing partnerships and supply chain management, BMW and other companies in the industry can drive further operational improvements and maintain a competitive advantage in the marketplace.

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